

SAMBP Technical Report TR-70/2028

Meshed IBR Topology Engine for SAMBP Protection

Multi-Source Network Solver, Pilot Schemes, and IBR Reach Adjustment: 88-Scenario Integration

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1 Background and Objectives

1.1 Motivation

Meshed transmission networks with high IBR penetration present a fundamental challenge for distance protection: the superposition principle no longer holds in the conventional sense because IBR sources inject fault current with magnitudes and angles that depend on pre-fault operating point, converter control mode (GFL vs GFM), and the FRT current injection strategy [1].

In a meshed topology with n IBR sources and m buses, the fault current seen by a distance relay at bus k for a fault at bus f is:

$$I_k^f = \sum_{j=1}^n \alpha_{kj}^f I_j^{\text{IBR}} + \sum_{l \in \mathcal{G}} \alpha_{kl}^f I_l^{\text{SG}} \quad (1)$$

where α_{kj}^f are network distribution factors dependent on the full admittance matrix $\mathbf{Y}_{\text{bus}}$ and the IBR current injection vector changes dynamically with converter control loops [2].

1.2 Objectives

1. Implement a multi-source network solver that builds $\mathbf{Z}_{\text{bus}}$ dynamically as IBR units trip or change mode.
2. Implement and validate three pilot protection schemes: POTT, DCB, and PUTT.
3. Develop the IBR reach adjustment formula: $\text{factor} = 1 - 0.10 f_{\text{gfm}} k_{\text{ibr}}$.
4. Integrate with the EKF tracker for multi-source cross-coupling estimation.
5. Validate 88 integration scenarios: 5-bus and 10-bus test networks.

2 Theory and Methodology

2.1 Multi-Source Network Solver

The solver builds the full $n \times n$ bus admittance matrix from:

- Line and cable admittances (from topology database).
- IBR Norton equivalents: $Y_{\text{ibr},j} = 1/Z_{\text{ibr},j}$ where $Z_{\text{ibr},j}$ depends on converter control bandwidth.
- SG Thevenin equivalents: $Z_{\text{sg},l} = jX_d''$.

Inverting the reduced $\mathbf{Y}_{\text{bus}}$ (Kron reduction for load buses) gives $\mathbf{Z}_{\text{bus}}$, from which the distribution factors in (1) are extracted: $\alpha_{kj}^f = (Z_{kk} - Z_{kf})/(Z_{ff} + Z_f^{\text{ext}})$.

2.2 Pilot Protection Schemes

POTT (Permissive Over-reaching Transfer Trip). Both ends of the line must detect the fault (forward over-reaching) before tripping. The permissive signal is communicated via GOOSE within 4 ms. For IBR-rich lines, the forward directional element uses positive-sequence current polarisation to avoid the negative-sequence ambiguity under GFM sources.

DCB (Directional Comparison Blocking). Any terminal that detects a fault in the reverse direction sends a block signal. In the absence of a block signal, the local forward element is permitted to trip after a coordination timer $t_{\text{DCB}} = 15$ ms. Blocking is robust to IBR current reversal because the directional element uses memorised voltage polarisation.

PUTT (Permissive Under-reaching Transfer Trip). The local Zone 1 element (under-reaching, 80% of line impedance) issues a permissive signal to the remote end. Zone 2 at the remote end then trips immediately upon receiving the permissive. PUTT avoids the reach ambiguity of POTT by relying only on Zone 1 (which is not susceptible to IBR infeed).

2.3 IBR Reach Adjustment Formula

The apparent impedance seen by a distance relay under IBR infeed is: $Z_{\text{app}} = Z_{\text{line}} \cdot (1 + k_{\text{ibr}})$, where k_{ibr} is the IBR current ratio (defined in TR-56). The GFM fraction f_{gfm} further modifies the apparent impedance because GFM sources maintain voltage during faults (reducing the infeed effect):

$$Z_{\text{reach}} = Z_{\text{set}} \times \underbrace{(1 - 0.10 f_{\text{gfm}} k_{\text{ibr}})}_{\text{reach adjustment factor}} \quad (2)$$

The factor 0.10 is calibrated from the 5-bus and 10-bus simulation campaigns (Section 3) to minimise over-reach risk while maintaining selectivity.

2.4 Multi-Source EKF with Cross-Coupling

When $n \geq 2$ IBR sources feed the same fault, their contributions are correlated through the network. The state vector of the multi-source EKF is:

$$\mathbf{x}_{\text{ms}} = [k_{\text{ibr}}^{(1)}, f_{\text{gfm}}^{(1)}, k_{\text{ibr}}^{(2)}, f_{\text{gfm}}^{(2)}, \dots]^T \quad (3)$$

with cross-coupling captured in the off-diagonal elements of the process noise covariance $\mathbf{Q}_{\text{ms}}$: $Q_{ij} = \rho_{ij} \sqrt{Q_{ii} Q_{jj}}$ where ρ_{ij} is the electrical distance correlation coefficient between sources i and j .

3 Simulation Results

3.1 Test Networks

Table 1: Test network configurations for TR-70.

Network	Buses	IBR Units	Pilot Scheme	Scenarios
5-bus ring	5	2 (GFL + GFM)	POTT, DCB	44
10-bus mesh	10	4 (2 GFL + 2 GFM)	POTT, DCB, PUTT	44
Total	—	6	—	88

3.2 88-Scenario Integration Results

Table 2: TR-70 88-scenario agreement results.

Test Group	Scenarios	AGREE	DISAGREE	Agree Rate	Status
T1: POTT (5-bus, SLG)	18	18	0	100%	✓
T2: DCB (5-bus, 3PH)	16	16	0	100%	✓
T3: POTT (10-bus)	18	18	0	100%	✓
T4: DCB (10-bus)	18	18	0	100%	✓
T5: PUTT (10-bus)	18	18	0	100%	✓
Total	88	88	0	100%	PASS

3.3 Reach Adjustment Validation

The reach adjustment factor $1 - 0.10 f_{\text{gfm}} k_{\text{ibr}}$ was tested across $f_{\text{gfm}} \in \{0, 0.5, 1.0\}$ and $k_{\text{ibr}} \in \{0.0, 0.5, 1.0\}$. Without the adjustment, the distance relay over-reaches by up to 12% when $f_{\text{gfm}} = 1.0$ and $k_{\text{ibr}} = 1.0$. With the adjustment, over-reach is reduced to $< 1\%$ across all 88 scenarios.

Table 3: Reach adjustment effectiveness ($k_{\text{ibr}} = 1.0$, varying f_{gfm}).

f_{gfm}	Over-reach (unadjusted)	Over-reach (adjusted)	Factor
0.0	0%	0%	1.000
0.5	5.8%	0.3%	0.950
1.0	11.5%	0.8%	0.900

4 Conclusion

Finding 1 (Meshed topology engine: 100% agreement). *The multi-source network solver with dynamic $\mathbf{Z}_{\text{bus}}$ and the IBR reach adjustment formula achieve 88/88 = 100% scenario agreement across both the 5-bus ring and 10-bus meshed test networks.*

Finding 2 (IBR reach adjustment formula). *The formula factor = $1 - 0.10 f_{\text{gfm}} k_{\text{ibr}}$ reduces distance relay over-reach from up to 12% (unadjusted) to < 1% (adjusted) across all GFM penetration levels tested.*

Finding 3 (PUTT preferred for IBR-rich meshed networks). *Among the three pilot schemes, PUTT provides the most reliable operation because it relies only on Zone 1 under-reaching elements, which are immune to IBR infeed effects.*

References

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